Domain-Adversarial Training Under Label Shift Can Silently Invert Out-of-Distribution Detection

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Editors: Under Review for MIDL 2027

Abstract

Domain-adversarial training (DANN) is a standard remedy for site and device shortcuts in medical imaging, and deployed models increasingly rely on out-of-distribution (OOD) monitors to flag unfamiliar inputs. We show that the two can conflict silently. On skin-lesion classification with a lesion-disjoint ISIC 2019 split, gradient reversal toward smartphone images drives Mahalanobis OOD detection from AUROC 0.825 to 0.381 (95% CI $[0.358, 0.404]$), and below chance on three clinical datasets no part of training saw (Fitzpatrick17k 0.405; MED-NODE and SD-198 [PENDING: last-checkpoint values]), while in-distribution balanced accuracy stays at 0.47–0.48 [PENDING: paired CI, calibration]. The inversion is caused by label shift between the domains the adversary aligns: changing only the class mix the adversary sees moves unseen-domain AUROC by 0.24–0.37, and the effect persists at matched domain leakage. Mechanistically, DANN pulls unfamiliar images toward the majority class; under label shift it reaches chance-level unweighted domain leakage by misaligning classes, while the domain stays decodable within each class. The consequence is clinical: the share of melanomas from unseen sources predicted benign nevus with confidence above 0.9 rises from 8% to 49–81%. Importance-weighted alignment helps in proportion to how well the target class priors are known: with label-free BBSE estimates it recovers 32–45% of the lost AUROC at matched leakage, and with priors from 50 labelled target images detection returns to its pre-adaptation level, at a cost of 0.08 in-distribution balanced accuracy. OOD monitoring should be re-validated after domain adaptation, and domain leakage should be measured within class.

Keywords: domain-adversarial training; out-of-distribution detection; label shift; calibration; skin lesion classification; deployment monitoring

1. Introduction

Skin lesion classifiers learn acquisition artefacts as readily as pathology: ink markings, rulers, dermatoscope type and contributing site all correlate with diagnosis in public datasets (Winkler et al., 2019; Bissoto et al., 2020; Daneshjou et al., 2022). The standard remedy is to make the representation invariant to the acquisition domain, most often by domain-adversarial training with a gradient reversal layer (Ganin et al., 2016; Bousmalis et al., 2016). A deployed model also needs a second safeguard: an out-of-distribution (OOD) monitor that flags inputs unlike its training data so that a clinician reviews them. This paper asks what invariance training does to that monitor.

In our setting it can turn the monitor upside down. As the adversarial weight grows, OOD detection does not merely weaken; it falls below chance, so that images from clinics the model has never seen score as *more* familiar than its own test data. Nothing in a standard evaluation reveals this: in-distribution accuracy and calibration stay flat, and the

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domain-leakage probe, the quantity the method is meant to reduce, improves [PENDING: ID accuracy/ECE at last checkpoint]. The cost falls on the patients a monitor exists to protect: under the same training, melanomas from unseen clinics are predicted benign nevus with confidence above 0.9 up to ten times more often, and the OOD monitor that should route them to a clinician ranks them as routine.

We trace the failure to label shift. ISIC 2019 is 50.8% nevus, the smartphone images the adversary aligns against are 12.3% nevus, and matching the marginal feature distributions of domains with different class priors forces the representation to misplace one domain's classes (Zhao et al., 2019; Tachet des Combes et al., 2020). We show the consequence for OOD monitoring with an intervention that changes only the class mix the adversary sees, and we locate the mechanism in the feature space.

Contributions.

1. **The failure.** Under a lesion-disjoint split, gradient reversal inverts Mahalanobis OOD detection on held-out smartphone images and on three clinical datasets that no branch of training saw, with flat in-distribution accuracy and calibration, under two training recipes whose accuracy is within 0.06 of a standard ERM baseline on the same split.

2. **The cause.** Setting the class mix seen by the adversary to that of the target removes the inversion on unseen domains (AUROC difference 0.24–0.37); a regression at matched domain leakage attributes the effect to label shift rather than to deeper invariance.

3. **The mechanism and a measurement lesson.** DANN pulls images that differ from the source along the domain direction toward the majority class. Under label shift, an unweighted domain probe reports chance-level leakage while a within-class probe still reads the domain: invariance should be measured within class.

4. **Consequence and mitigation.** Melanomas from unseen sources are called benign nevi with high confidence up to ten times more often. Importance weighting with label-free prior estimates recovers part of the loss; with priors from 50 labelled target images it recovers fully, at a cost in in-distribution accuracy.

2. Related work

Shortcuts and invariance in dermatology. Ink markings shift melanoma predictions (Winkler et al., 2019), classifiers keep competitive ISIC accuracy with the lesion masked out (Bissoto et al., 2020), and performance degrades on darker skin (Daneshjou et al., 2022). Domain-adversarial training (Ganin et al., 2016) and domain separation networks (Bousmalis et al., 2016) are the usual remedies; our dual-encoder model is a domain separation network and is a vehicle for the intervention, not a contribution.

Invariance under label shift. When class priors differ between domains, matching marginal feature distributions cannot also keep the joint error small (Zhao et al., 2019); importance-weighted alignment (IWDAN, IWCDAN) corrects for this with estimated target class priors (Tachet des Combes et al., 2020), which can be estimated without target

labels by black-box shift estimation (Lipton et al., 2018) or EM (Saerens et al., 2002). These results concern classification error. We study what the same misalignment does to OOD monitoring, and show that a standard domain-leakage probe misreports invariance under label shift.

OOD detection after adaptation. Full-spectrum OOD detection (Yang et al., 2023; Zhang et al., 2023) asks that covariate-shifted in-distribution inputs be accepted; our smartphone and clinical images fall in that cell, and Section 5 argues that acceptance is safe only when the model also classifies them. Calibration degrades under shift (Ovadia et al., 2019) and after domain adaptation (Wang et al., 2020). Closest to our observation, Wu et al. report Mahalanobis and ODIN AUROC below 0.5 on rotated MNIST after adaptation with an MMD-based method (DAN) (Wu et al., 2023), as a baseline in a study of continually adaptive detection. We show the inversion on medical images, identify label shift as its cause, and characterise the mechanism.

3. Materials and methods

3.1. Data and splits

ISIC 2019 (Tschandl et al., 2018) (25,331 dermoscopic images, eight classes: MEL, NV, BCC, AK, BKL, DF, VASC, SCC) is the in-distribution source. We split it 60/20/20 into train, validation and test *grouped by* `lesion_id` (16,211 / 4,053 / 5,067 images), because 60% of test images in the image-level split share a lesion with training (Tran, 2027). **PAD-UFES-20** (Pacheco et al., 2020) (2,298 smartphone photographs, six of the eight classes) is the domain the adversary aligns against. Its labels are never used in any loss. We split it by patient into `pad_adv` (1,582 images, 961 patients), seen by the adversarial and context branches, and `pad_heldout` (716 images, 412 patients), never seen.

Unseen domains. Four clinical-photograph datasets enter no part of training: Fitzpatrick17k (Groh et al., 2021) (3,887 images obtained), MED-NODE (170 images), SD-198 (6,584 images) and SCIN (10,406 images, 5,033 cases) [PENDING: cite MED-NODE, SD-198, SCIN; licences: SCIN CC BY 4.0, MED-NODE none stated, SD-198 official source]. SCIN is mostly inflammatory conditions; only 0.6% of its images map to the ISIC classes, so we use it only as an OOD set for detection and report it separately; no label-based analysis is done on SCIN. Images are cropped to a lesion-centred box and resized to 224×224.

3.2. Model and training

A dual-encoder model with EfficientNet-B3 backbones produces a lesion latent $z_\ell \in \mathbb{R}^{16}$ (batch-normalised, z_ℓ^{norm}) used for diagnosis through a linear head, and a context latent $z_c \in \mathbb{R}^{64}$. The objective is

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{ctx}}\mathcal{L}_{\text{ctx}} + \lambda_{\text{adv}}\,\mathcal{L}_{\text{adv}} + \lambda_{\text{orth}}\mathcal{L}_{\text{orth}}, \tag{1}$$

with $\mathcal{L}_{\text{cls}}$ on ISIC only, $\mathcal{L}_{\text{ctx}}$ a domain cross-entropy on z_c, $\mathcal{L}_{\text{adv}}$ a domain cross-entropy on z_ℓ^{norm} behind a gradient reversal layer, and $\mathcal{L}_{\text{orth}}$ a squared-cosine penalty between the latents. AdamW (learning rate 10^{-4}; adversary $\times 30$, classifier $\times 0.2$), 40 epochs, domain-balanced batches. We sweep $\lambda_{\text{adv}} \in \{0, 0.25, 0.5, 1, 2, 4, 8\}$ with everything else fixed (seeds

42–82 at $\lambda_{\text{adv}} \in \{0, 2\}$, 42–46 elsewhere). A second recipe (A4: $z_\ell \in \mathbb{R}^{64}$, classifier $\times 1$, class-weighted loss) serves as a robustness check.

Checkpoints. We report the last checkpoint. Validation balanced accuracy (eight classes, chance 0.125) has a lesion-clustered bootstrap SE of 0.023–0.026, and the best-validation and last checkpoints differ by less than one SE, so best-validation selection picks early epochs by noise. Best-validation values are given for comparison.

Competence. In-distribution balanced accuracy at $\lambda_{\text{adv}} = 0$ is 0.496 (main recipe) and 0.520 (A4), 0.04–0.06 below a standard single-encoder ERM on the same split (0.556 [PENDING: ERM reference at 5 seeds, with SE]) and four times chance (0.125). Lesion-disjoint eight-class ISIC 2019 is hard; the image-level split inflates the same model's balanced accuracy by 0.20 (Tran, 2027).

3.3. Interventions on label shift

Class mix seen by the adversary. Keeping the classification loss on all of ISIC, we resample only the ISIC images fed to the adversarial branch so that their class mix is ISIC's own (100%, the default), halfway to PAD's (50%), or PAD's (0%, no label shift between the aligned domains), at $\lambda_{\text{adv}} \in \{0.5, 2\}$ with five seeds (three at 50%).

Importance-weighted alignment. IWDAN and IWCDAN (Tachet des Combes et al., 2020) reweight the source in the adversarial loss by estimated target class priors; we compare BBSE estimates (Lipton et al., 2018), EM with temperature-scaled probabilities (Saerens et al., 2002), priors from 50 labelled `pad_adv` images (a few-label setting), and the true `pad_adv` priors (oracle, not deployable). CDAN (Long et al., 2018) is a class-conditional adversarial baseline.

3.4. Measurements

OOD detection. Class-conditional Mahalanobis with shared covariance (Lee et al., 2018) (primary), k-NN (Sun et al., 2022), maximum cosine to a class centroid, maximum softmax probability (Hendrycks and Gimpel, 2017) and energy (Liu et al., 2020), all fit on ISIC train. AUROC treats OOD as the positive class, so 0.5 is chance and values below 0.5 mean OOD images score as more familiar than ID test images. The five scores are functions of the same latent, so they are not independent evidence.

Inversion. We call a cell inverted when the cluster-bootstrap 95% CI of the seed-mean Mahalanobis AUROC lies below 0.5 and at least three seeds' own intervals do (all three in cells with three seeds). Bootstrap draws (2,000) resample ISIC test by lesion and OOD sets by patient (`pad_heldout`), near-duplicate cluster (Fitzpatrick17k) or case (SCIN); MED-NODE and SD-198 carry no patient identifiers and are resampled by image, so their intervals may be too narrow. Runs with in-distribution balanced accuracy below 0.25 are excluded as degraded and reported.

Domain leakage. Balanced accuracy (chance 0.5) of a logistic probe for the source dataset on frozen z_ℓ^{norm}. The probe used in earlier work scores ISIC test against all of PAD, including images the adversary trained on. We therefore also report probes on held-out

Table 1: The adversarial-weight sweep (main recipe, lesion-disjoint ISIC split, last checkpoint). Mean ± s.d. over five seeds per λ_{adv} (42–82 at $\lambda_{\text{adv}} \in \{0, 2\}$, 42–46 otherwise). Leakage: linear domain probe on held-out images (chance 0.5). ID: balanced accuracy, eight classes (chance 0.125). Mahalanobis AUROC, OOD positive; k: seeds whose interval lies below 0.5. Best-val: the best-validation checkpoint, for comparison.

λ_{adv}	leakage	ID bal. acc	`pad_heldout`	k	Fitzpatrick17k	k	`pad_heldout` (best-val)
0	0.842 ± 0.050	0.481 ± 0.016	0.825 ± 0.022	0/5	0.712 ± 0.013	0/5	0.808 ± 0.027
0.25	0.502 ± 0.002	0.481 ± 0.010	$\mathbf{0.445 \pm 0.022}$	4/5	$\mathbf{0.396 \pm 0.032}$	5/5	0.525 ± 0.121
0.5	0.508 ± 0.010	0.484 ± 0.005	$\mathbf{0.415 \pm 0.016}$	5/5	$\mathbf{0.377 \pm 0.046}$	5/5	0.426 ± 0.008
1	0.505 ± 0.005	0.482 ± 0.017	$\mathbf{0.390 \pm 0.022}$	5/5	$\mathbf{0.388 \pm 0.039}$	5/5	0.387 ± 0.016
2	0.507 ± 0.009	0.472 ± 0.018	$\mathbf{0.381 \pm 0.023}$	5/5	$\mathbf{0.405 \pm 0.031}$	5/5	0.403 ± 0.028
4	0.527 ± 0.035	0.480 ± 0.016	$\mathbf{0.416 \pm 0.034}$	5/5	$\mathbf{0.354 \pm 0.066}$	5/5	0.433 ± 0.020
8	0.520 ± 0.028	0.480 ± 0.018	$\mathbf{0.439 \pm 0.020}$	5/5	$\mathbf{0.383 \pm 0.036}$	5/5	0.454 ± 0.044

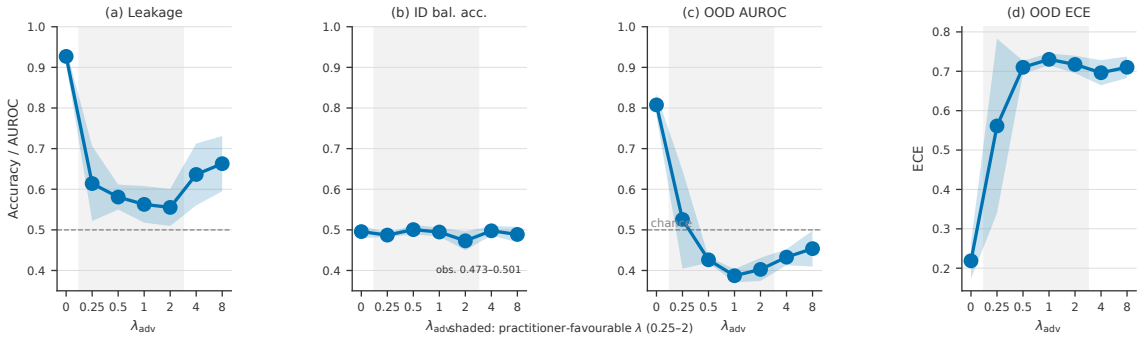


Figure 1: **The intervention, four views, one axis.** Leakage falls to its chance floor and in-distribution balanced accuracy does not degrade — both favourable. Over the same range OOD AUROC collapses through chance and inverts, and out-of-domain calibration error triples. A developer monitoring the first two panels would see nothing wrong.

images only (linear, one-hidden-layer MLP, k-NN), a class-balanced probe that reweights both domains to a common class mix, and a within-class probe averaged over classes.

Calibration and diagnosis. Expected calibration error (15 bins) on ISIC test and on `pad_heldout` (six shared classes), and zero-shot balanced accuracy on `pad_heldout`. Out-of-domain ECE is close to mean confidence minus accuracy here, so these are views of one phenomenon.

Table 2: Unseen domains: Mahalanobis AUROC at $\lambda_{\text{adv}} = 0$ and $\lambda_{\text{adv}} = 2$, mean ± s.d. over five seeds, with the number of seeds whose interval lies below 0.5. Main recipe, best-validation checkpoint [PENDING: MED-NODE/SD-198/SCIN at the last checkpoint]; A4, last checkpoint. MED-NODE and SD-198 are resampled by image. –: not bootstrapped.

recipe	set	$\lambda_{\text{adv}} = 0$	$\lambda_{\text{adv}} = 2$	seeds < 0.5	inverted
main	`pad_heldout`	0.808 ± 0.027	0.403 ± 0.028	5/5	yes
main	Fitzpatrick17k	0.717 ± 0.034	0.402 ± 0.018	5/5	yes
main	MED-NODE	0.536 ± 0.037	0.306 ± 0.071	5/5	yes
main	SD-198	0.713 ± 0.037	0.375 ± 0.026	5/5	yes [PENDING: SD-198 cluster CI]
main	SCIN	0.726 ± 0.021	0.559 ± 0.034	0/5	no
A4	`pad_heldout`	0.786 ± 0.019	0.452 ± 0.014	[PENDING: k]	yes
A4	Fitzpatrick17k	0.821 ± 0.023	0.458 ± 0.061	[PENDING: k]	yes

4. Results

4.1. OOD detection inverts while in-distribution metrics stay flat

The sweep. At the last checkpoint, Mahalanobis AUROC on `pad_heldout` falls from 0.825 at $\lambda_{\text{adv}} = 0$ to 0.445 at $\lambda_{\text{adv}} = 0.25$ and stays between 0.381 and 0.445 at every larger weight, meeting the inversion criterion at all six non-zero weights; on Fitzpatrick17k it falls from 0.712 to 0.354–0.405 (Table 1). At $\lambda_{\text{adv}} = 2$ the seed-mean interval is $[0.358, 0.404]$ on `pad_heldout` and $[0.392, 0.419]$ on Fitzpatrick17k. In-distribution balanced accuracy stays within 0.472–0.484 [PENDING: paired $\lambda_{\text{adv}} = 2 - \lambda_{\text{adv}} = 0$ CI and ID/OOD ECE at the last checkpoint]. All five scores fall below 0.5 on `pad_heldout` at $\lambda_{\text{adv}} = 2$ (k-NN 0.427, cosine 0.420, MSP 0.455, energy 0.468; best-validation checkpoint); the standard k-th-neighbour k-NN gives the same values within 0.01. The best-validation checkpoint gives the same picture from $\lambda_{\text{adv}} = 0.5$ on, but at $\lambda_{\text{adv}} = 0.25$ it selects an earlier epoch and the inversion is not yet visible (0.525, seed s.d. 0.121; an observation made after the sweep).

Unseen domains. The inversion is not confined to the domain the adversary was trained against (Table 2). On three clinical datasets that entered no part of training, Mahalanobis AUROC falls from 0.717 to 0.402 (Fitzpatrick17k), 0.536 to 0.306 (MED-NODE) and 0.713 to 0.375 (SD-198) at $\lambda_{\text{adv}} = 2$, each meeting the inversion criterion. On SCIN, whose images are mostly inflammatory conditions outside the ISIC label space, detection falls from 0.726 to 0.559 without inverting.

A second recipe. With a 64-dimensional latent, unweighted cross-entropy replaced by a class-weighted loss and the last checkpoint (A4), detection again inverts at $\lambda_{\text{adv}} = 0.5$ and $\lambda_{\text{adv}} = 2$ on `pad_heldout` (0.468 $[0.440, 0.495]$; 0.452 $[0.425, 0.478]$) and Fitzpatrick17k (0.444; 0.458), with in-distribution balanced accuracy unchanged (0.520, 0.520, 0.510). On `pad_heldout` all five scores' intervals lie below 0.5; on Fitzpatrick17k k-NN (0.521) and energy (0.516) only collapse. A plain single-encoder DANN collapses to chance (0.51–0.56) without inverting (Appendix A.3).

Table 3: Label-shift intervention: Mahalanobis AUROC (main recipe, last checkpoint) by the class mix the adversary sees. Shift 100%: ISIC's own mix; 0%: PAD's mix. Leakage: probe of Table 1. Shift 0% at $\lambda_{\text{adv}} = 2$: four seeds after excluding one degraded run (in-distribution balanced accuracy $0.139 < 0.25$); shift 50%: three seeds.

λ_{adv}	shift	leakage	`pad_heldout`	Fitzpatrick17k	MED-NODE	SD-198
0	–	0.928	0.825	0.712	[PENDING:]	[PENDING:]
0.5	0%	0.790	0.499	0.674	[PENDING:]	[PENDING:]
0.5	50%	0.629	0.490	0.482	[PENDING:]	[PENDING:]
0.5	100%	0.628	0.361	0.385	[PENDING:]	[PENDING:]
2	0%	0.791	0.458	0.691	0.541	0.694
2	50%	0.658	0.454	0.441	[PENDING:]	[PENDING:]
2	100%	0.634	0.368	0.340	0.298	0.325

4.2. Label shift between the aligned domains causes the inversion

Intervention. Changing only the class mix of the ISIC images the adversary sees switches the inversion on unseen domains on and off (Table 3). At $\lambda_{\text{adv}} = 2$, with the adversary's ISIC images resampled to PAD's class mix (shift 0%), Mahalanobis AUROC is 0.691 on Fitzpatrick17k, 0.541 on MED-NODE and 0.694 on SD-198, none inverted; with ISIC's own mix (shift 100%) it is 0.340, 0.298 and 0.325, inverted in every seed. Paired by seed, removing the label shift raises AUROC by 0.361, 0.243 and 0.372. The intermediate level falls between the two (Fitzpatrick17k 0.482 at $\lambda_{\text{adv}} = 0.5$, 0.441 at $\lambda_{\text{adv}} = 2$). On `pad_heldout`, the adversary's own target domain, a mild inversion remains without label shift (0.458; paired difference +0.093). One of five shift-0% runs collapsed (in-distribution balanced accuracy 0.139) and is excluded.

Matched leakage. Removing label shift also leaves more domain information in the latent (leakage 0.791 vs 0.634), so the intervention alone does not separate label shift from depth of invariance. A regression of unseen-domain AUROC on label-shift level and leakage, with recipe as a covariate (variance inflation ≈ 1.04), gives label-shift coefficients of -0.166 $[-0.217, -0.115]$ (Fitzpatrick17k), -0.107 $[-0.148, -0.065]$ (MED-NODE) and -0.172 $[-0.221, -0.123]$ (SD-198) for a change from no shift to full shift at fixed leakage [PENDING: confirm shift coding and resampling unit]. On `pad_heldout` the coefficient is $+0.041$ $[-0.033, +0.110]$. Comparing runs at equal leakage across recipes (A4 at shift 100%, leakage 0.78, against the main recipe at shift 0%, leakage 0.79) gives the same direction (+0.23 to +0.26).

A natural example without DANN. Label shift alone can invert a class-conditional Mahalanobis score. Training on BCN and treating HAM as OOD, Mahalanobis AUROC is already 0.369 $[0.336, 0.401]$ at $\lambda_{\text{adv}} = 0$, while k-NN gives 0.658. HAM-heldout is 67% nevus against 31% in BCN test; reweighting HAM to BCN's class mix gives 0.489 $[0.453, 0.524]$, and no class-matched AUROC differs from 0.5 (post hoc). In the main setting the OOD sets are *less* nevus-heavy than ISIC, so detection works at $\lambda_{\text{adv}} = 0$; DANN creates the condition by moving them toward nevus.

4.3. Mechanism: misaligned classes behind chance-level leakage

Unfamiliar images move to the majority class. At $\lambda_{\text{adv}} = 2$ with full label shift, the share of `pad_heldout` images whose nearest class centroid is nevus rises from 0.28 to 0.63, and to 0.75–0.85 on the unseen sets; without label shift it stays at 0.12 and 0.39–0.49. Images of non-nevus classes are pulled into the nevus region: split by true class, non-nevus OOD images fall to AUROC 0.42 (main recipe) and 0.47 (A4), whereas images still nearest a non-nevus centroid stay at 0.62 and 0.64. Because pooled AUROC is the class-share-weighted mean of per-class AUROCs, and nevus is half the ISIC test set, this displacement is enough to push the pooled score below chance.

The displacement follows the domain direction. For each OOD image we measured how far it lies along the direction separating ISIC from PAD in the $\lambda_{\text{adv}} = 0$ model, and how much closer to its nearest ID centroid it moves at $\lambda_{\text{adv}} = 2$. The two correlate on every lesion dataset (Spearman 0.33–0.47), while the deviation orthogonal to that direction does not predict the pull ($|\rho| \leq 0.07$ on four of five sets). Between two $\lambda_{\text{adv}} = 0$ models of different seeds the same correlation is smaller (0.11–0.29); the DANN-specific excess exceeds 0.15 on all four unseen datasets (0.18–0.33) but not on `pad_heldout` (0.057), and is larger in the pre-projection backbone (0.24–0.45 on all five).

Chance-level leakage hides the misalignment. The domain probe of Table 1 is scored partly on images the adversary trained on. On held-out images a linear probe reads the domain at chance from $\lambda_{\text{adv}} = 0.5$ on (0.502–0.506), but a k-NN probe still reads it (0.55–0.61), and under label shift an unweighted probe misreports invariance (Table 4). With full label shift DANN reaches 0.510 on an unweighted held-out probe, yet within each class the domain remains decodable (0.618), more so than in any other condition. Correcting the class mix the adversary aligns (shift 0%, or importance weighting with the true priors) raises the unweighted reading (0.63–0.64) while lowering the within-class one (0.56–0.57). Under label shift, DANN matches the marginal distributions by misaligning classes, as the theory of Zhao et al. (2019) implies; leakage should therefore be measured within class (post hoc).

4.4. Clinical consequence and mitigation

Melanomas called benign. With a threshold accepting 95% of ISIC validation images, almost all unseen-domain malignant images are accepted at every λ_{adv}, so the monitor does not catch them in either case. What changes is the classifier: the share of melanomas predicted nevus with confidence above 0.9 rises from 7.5% to 48.7% on Fitzpatrick17k and from 8.1% to 81.2% on SD-198 between $\lambda_{\text{adv}} = 0$ and $\lambda_{\text{adv}} = 2$ [PENDING: n and CIs per cell; label map].

No fix at scoring time. Reversing the sign of the score is not a remedy: at $\lambda_{\text{adv}} \geq 0.5$ it reaches only 0.55–0.63 on `pad_heldout` and Fitzpatrick17k with a false-positive rate of 0.84–0.95 at 95% true-positive rate [PENDING: verify FPR range in s_flip], and at $\lambda_{\text{adv}} = 0$ it would turn 0.808 into 0.192. Relative Mahalanobis (Ren et al., 2021) and per-class standardised distances stay below 0.5 on all three unseen datasets at $\lambda_{\text{adv}} = 2$.

Table 4: Domain leakage on held-out images under three probes, and unseen-domain detection ($\lambda_{\text{adv}} = 2$, main recipe, last checkpoint). Unweighted: as usually reported; class-balanced: both domains reweighted to a common class mix; within-class: class-size-weighted mean of per-class probes. Fitz: Mahalanobis AUROC on Fitzpatrick17k.

condition	unweighted	class-balanced	within-class	Fitz
DANN, shift 100%	0.510	0.602	0.618	0.340
DANN, shift 0%	0.629	0.539	0.560	0.691
IWDAN (BBSE)	0.534	0.562	0.597	0.508
IWDAN (50 labels)	0.653	0.545	0.583	0.741
IWDAN (true priors)	0.645	0.549	0.576	0.717

Importance-weighted alignment. IWDAN reweights the source in the adversarial loss by estimated target class priors, and how well it works depends on the estimate. With weights updated online from unlabelled target predictions, as in the original method, the implied target priors end 0.43–0.55 (total variation) from the truth: nevus is down-weighted from 0.51 toward 0.23–0.28, against a true 0.10. IWDAN then keeps leakage at the DANN level (0.642 vs 0.634) and recovers 32% of the lost AUROC on `pad_heldout` (0.513) and 45% on Fitzpatrick17k (0.508), while MED-NODE (0.350) and SD-198 (0.488) stay at or below chance. *EM* estimation was unstable: in two of five seeds it placed mass on classes absent from PAD, so we treat it as not having run correctly. With priors from 50 *labelled target images* (total variation 0.10–0.14), detection on the unseen domains returns to the level obtained with the true priors (Fitzpatrick17k 0.741 vs 0.717, SD-198 0.734 vs 0.693, MED-NODE 0.606 vs 0.563; `pad_heldout` 0.619 vs 0.647), and within-class leakage is lower than under DANN (0.58 vs 0.62). This is a promising but post hoc mitigation: neither the 50-label nor the true-prior variant meets our leakage criterion, because unweighted leakage rises (0.81 and 0.78 against 0.63), and both cost in-distribution balanced accuracy (0.39–0.40, against 0.48 both under DANN and before adaptation). Read descriptively, detection improves as the prior error falls (0.52: partial; 0.10–0.14: oracle level; 0: oracle), but our planned rank-correlation test across estimators was not supported (Spearman -0.33, $[-0.80, +0.28]$), with EM contributing invalid priors. Black-box shift estimates computed after training were further from the truth (0.65–0.76), plausibly because their assumption of unchanged class-conditional inputs fails across these domains [PENDING: IWCDAN, CDAN, $\lambda_{\text{adv}} = 0.5$ cells; per-method details to appendix].

5. Discussion

Tolerance without generalisation. Full-spectrum OOD detection asks that covariate-shifted inputs be accepted (Yang et al., 2023). After DANN our models accept unseen-clinic images more readily than their own test data while classifying them at a zero-shot balanced accuracy of about 0.25 (chance 0.167). A tolerant detector would score such inputs near 0.5; scoring them below 0.5 is not tolerance but a monitor that has lost its meaning. Claims of tolerance to covariate shift should report cross-domain accuracy on the same inputs.

Why label shift. Matching marginal feature distributions across domains with different class priors cannot keep both domains' classes in place (Zhao et al., 2019). Our results show what this misplacement does to a distance-based monitor: unfamiliar images that differ along the domain direction land in the majority class's region, where a class-conditional score reads them as typical. The BCN→HAM case shows that a nevus-heavy OOD set alone is enough to invert such a score; DANN produces that condition in feature space when the aligned domains differ in class mix. Without label shift the inversion stays on the adversary's own target domain, where a smaller within-class pull remains. The same misplacement explains why an unweighted domain probe reports chance-level leakage: it measures marginal, not class-conditional, alignment.

Checkpoint selection can hide the failure. As an observation made after the sweep, at $\lambda_{\text{adv}} = 0.25$ the last checkpoint is inverted on both `pad_heldout` and Fitzpatrick17k, whereas best-validation selection picks earlier epochs at which the seeds still disagree (seed s.d. 0.121). Validation accuracy cannot tell these checkpoints apart, so a model selected in the usual way can look safe because it was stopped before the failure formed.

No fix at scoring time. Reversing the score works only if one knows in advance that the model is inverted; at $\lambda_{\text{adv}} = 0$ it would turn a working monitor into a broken one, and at $\lambda_{\text{adv}} \geq 0.5$ it leaves most OOD images accepted at a 95% ID-acceptance threshold. Scores designed to be less sensitive to class structure, such as Relative Mahalanobis, also stay below chance. The remedy has to act on training, through the class priors the adversary aligns.

Recommendations. After any domain-alignment step: (i) re-validate OOD detection on images from unseen sources, per class as well as pooled; (ii) report domain leakage within class, not only unweighted; (iii) compare class priors between the aligned domains, and if they differ, correct for them, which in our setting needed a small labelled sample from the target domain; and (iv) report results at more than one checkpoint.

6. Limitations

Scope. The inversion is shown in dermatology only. We attempted two other modalities and neither experiment took place: on Camelyon17 no adversarial setting we tried removed hospital information before the classifier degraded, and on chest radiographs the detector was near chance on the OOD set before any adaptation (AUROC 0.522), so there was nothing to invert. **Architecture.** The main results use a dual-encoder model; the A4 recipe changes the latent size and loss, but a plain single-encoder DANN collapses detection to chance without inverting, and whether deeper invariance would invert it is untested [PENDING: R1-hi / P9]. **Competence.** In-distribution balanced accuracy is 0.04–0.06 below a standard ERM on the same lesion-disjoint split [PENDING: ERM at 5 seeds]. **Statistics.** MED-NODE and SD-198 carry no patient identifiers and are resampled by image, so their intervals may be too narrow; SCIN is used only as an OOD set because almost none of it maps to the ISIC classes. The five OOD scores are functions of one latent and are not independent evidence. **Mechanism.** A two-domain Gaussian toy model with label shift did not reproduce the inversion, so the conditions we identify may not be sufficient in general. **Warning signals.** No statistic computed on in-distribution data alone flagged

the inverted runs; the share of deployment images predicted as the majority class tracked inversion (Spearman -0.73) but needs unlabelled target data. **Mitigation.** The few-label remedy is a post hoc analysis, costs in-distribution accuracy, and did not meet our leakage criterion. **Data.** We obtained 3,887 of 16,577 Fitzpatrick17k images.

7. Conclusion

Domain-adversarial training can make an OOD monitor rank images from unseen clinics as more familiar than its own test data, while every routinely reported in-distribution metric stays flat. In our setting the cause is label shift between the domains the adversary aligns: it misplaces unfamiliar images into the majority class and hides the misalignment from the usual leakage probe. Checking OOD detection on unseen sources after adaptation, measuring leakage within class and correcting for class priors are inexpensive steps that make the failure visible and, with a small labelled target sample, reduce it.

Data and code availability

https://github.com/TranDucVinh-20225538/CSG-SKin. ISIC 2019 CC BY-NC 4.0; PAD-UFES-20 CC BY 4.0; Fitzpatrick17k annotations CC BY-NC-SA 3.0 with images used for evaluation only and not redistributed.

Declaration of competing interest

The authors declare no competing interests.

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Appendix A. Supporting tables and figures

A.1. The image-level split

Table 8 is the same sweep under the image-level ISIC split used by most published work on this dataset, in which 60.0% of test images share a `lesion_id` with training. Every pattern is the same; the in-distribution accuracies are about 0.20 higher throughout.

A.2. Confidence, accuracy and calibration on unseen domains

At $\lambda_{\text{adv}} = 0$ the model is appropriately hesitant on unfamiliar data: mean maximum-softmax confidence is 0.915 on the ISIC test split and 0.489 on `pad_heldout`. By $\lambda_{\text{adv}} = 0.25$ that gap has closed (0.913 versus 0.898), and by $\lambda_{\text{adv}} = 2$ it has inverted (0.930 versus 0.910).

Figure 6: same-modality site shift

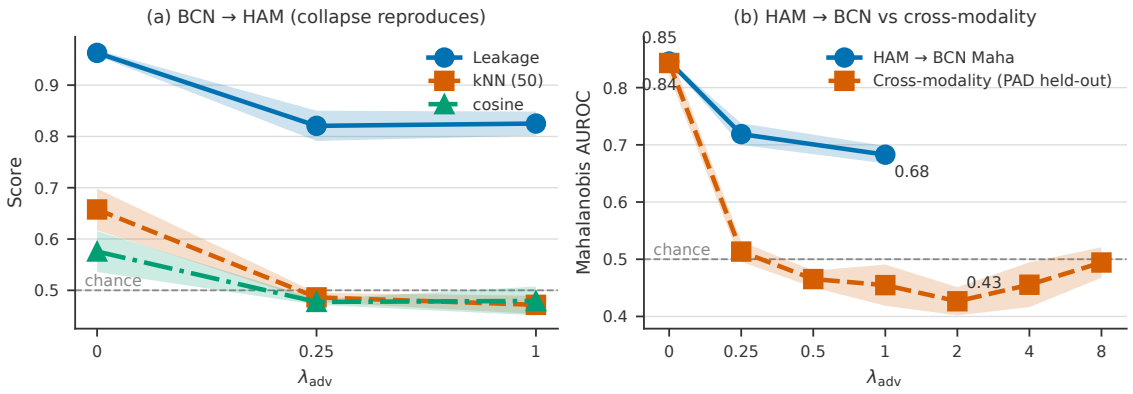


Figure 2: **A same-modality site shift separates the two claims.** (a) Barcelona to HAM10000: leakage falls and the detectors with headroom cross chance. (b) With the roles reversed to restore headroom, Mahalanobis falls from a baseline matching the cross-modality sweep but stops above chance, while the cross-modality sweep continues below it.

Cross-domain diagnostic accuracy does not improve at any λ_{adv}: balanced accuracy spans 0.249–0.291 against a six-class floor of 0.167, no setting exceeds the 0.291 obtained at $\lambda_{\text{adv}} = 0$, and every setting is below the 0.298 of the non-adversarial EfficientNet-B3 control. Plain accuracy falls 0.245 → 0.184, beneath the 0.385 majority-class rate throughout. Macro AUC is roughly preserved (0.605 → 0.581), so discrimination survives while the decision rule does not.

Predictions collapse toward the benign majority class: nevus predictions rise from 113 to 374 of 716 images, and actinic keratosis recall falls from 0.245 to 0.016 (49/202 → 3/202 images, mean over five seeds). Squamous cell carcinoma moves the same way (0.088 → 0.021, 6/66 → 1/66) but the count is too small to carry a claim on its own, and melanoma ($n = 9$) supports none.

In this setting out-of-domain ECE is arithmetically close to mean confidence minus accuracy, which is why the confidence, accuracy and calibration results are not independent corroboration of each other.

A.3. Single-encoder replication

Plain single-encoder domain-adversarial training on the same data and splits. Every detector collapses at $\lambda_{\text{adv}} = 0.25$ — k-NN 0.944 → 0.528, energy 0.911 → 0.507, cosine 0.884 → 0.509, MSP 0.868 → 0.515 — and by $\lambda_{\text{adv}} = 4$ four of the five sit below chance, cosine most clearly at 0.454 ± 0.014. In-distribution accuracy is flat throughout (0.845 → 0.832) and cross-domain accuracy falls from 0.390 to a plateau near 0.32.

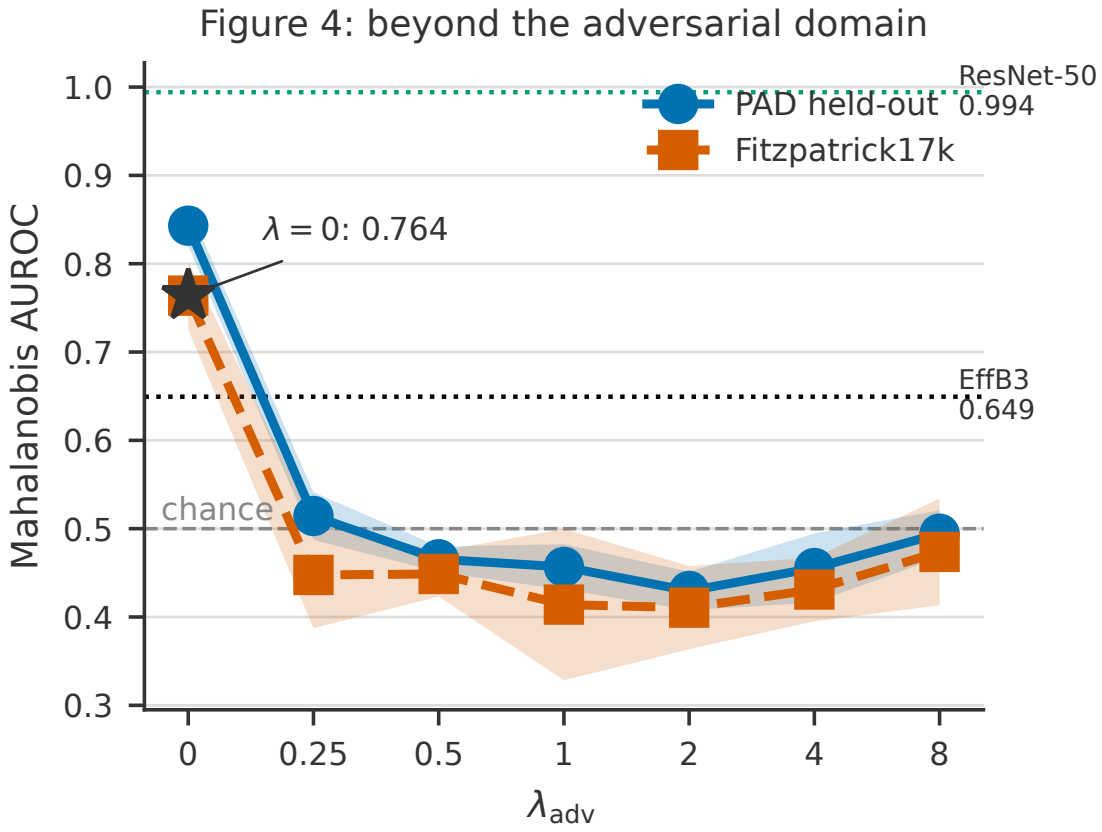


Figure 3: **The collapse is not confined to the adversarial domain.** AUROC against λ_{adv} for the held-out PAD partition and for Fitzpatrick17k, which is absent from every training branch. Horizontal references give the ResNet-50 baseline and the EfficientNet-B3 control on Fitzpatrick; the controlled comparison is the $\lambda_{\text{adv}} = 0$ point of the sweep itself.

A.4. External replication attempts

Camelyon17, three hospitals. The adversary head never trained: cross-entropy sat at $\ln 3$ from the first epoch at every λ_{adv} while a post-hoc probe on the same features read 0.94, despite a gradient graph verified correct (head cosine $+1.000$, encoder -1.000).

iWildCam. The adversary did train — best-epoch cross-entropy 3.13 against $\ln 243 = 5.49$ — but the encoder never became invariant: the binary train-versus-OOD site probe moved only from 0.911 to 0.891 across $\lambda_{\text{adv}} \in [0, 10]$ against a floor of 0.5, and no detector collapsed.

Camelyon17, two hospitals. Pre-committed in `PRECOMMIT_CAMELYON2.json` before running. The adversary reached chance (final cross-entropy 0.692 against $\ln 2 = 0.693$, from 0.11 at $\lambda_{\text{adv}} = 0$) and in-distribution accuracy held at 0.995, but the hospital probe fell only $0.959 \pm 0.007 \rightarrow 0.952 \pm 0.001$, inside its own seed spread, so the leakage gate

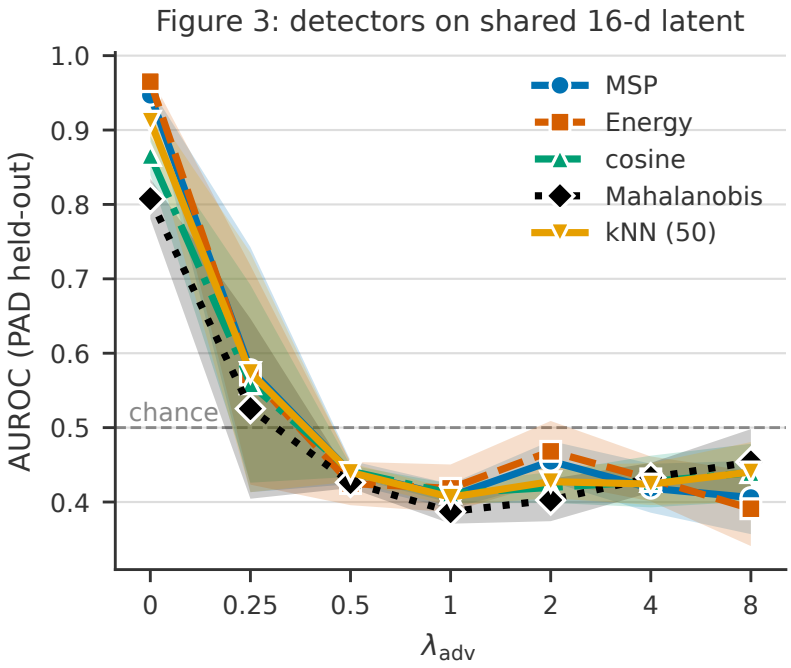


Figure 4: **Every scoring rule moves together.** The five detectors against λ_{adv} on the held-out PAD partition, mean ± 1 s.d. All five are functions of the same 16-dimensional latent and are therefore not independent evidence; Figure 6 carries the replication on a different representation.

failed and the verdict is negative. The exploratory seeds added afterwards do not reopen it: Mahalanobis on the held-out hospital reads 0.534 ± 0.050 at $\lambda_{adv} = 1$ over seeds 42–44, above chance, so the single-seed 0.478 did not replicate. These runs appear in Section 4.3 for the non-mediation claim only, where the five detectors move 0.12–0.22 while the probe moves 0.007.

A.5. Same-modality site shift: probe validation and the HAM direction

The sites are separable, and not for trivial reasons. A logistic probe on frozen ERM features distinguishes BCN from HAM at balanced accuracy 0.962 ± 0.007 against a floor of 0.5. Masking the dermatoscope vignette reduces this only to 0.943 ± 0.005, so the border is not the signal. Hand-crafted colour and intensity statistics reach 0.901 at best — close to the 0.876 ceiling the same features reach on ISIC↔PAD, so this pair is no more low-level than the original contrast — while sharpness statistics sit at chance (0.532). A probe given only the ground-truth class label reaches 0.668, reflecting the two sites' different class distributions; the feature probe is far above that reference.

Mahalanobis is uninformative in the BCN-in-distribution direction. It reads 0.379 at $\lambda_{adv} = 0$, already below chance before any adversarial training. Reweighting the

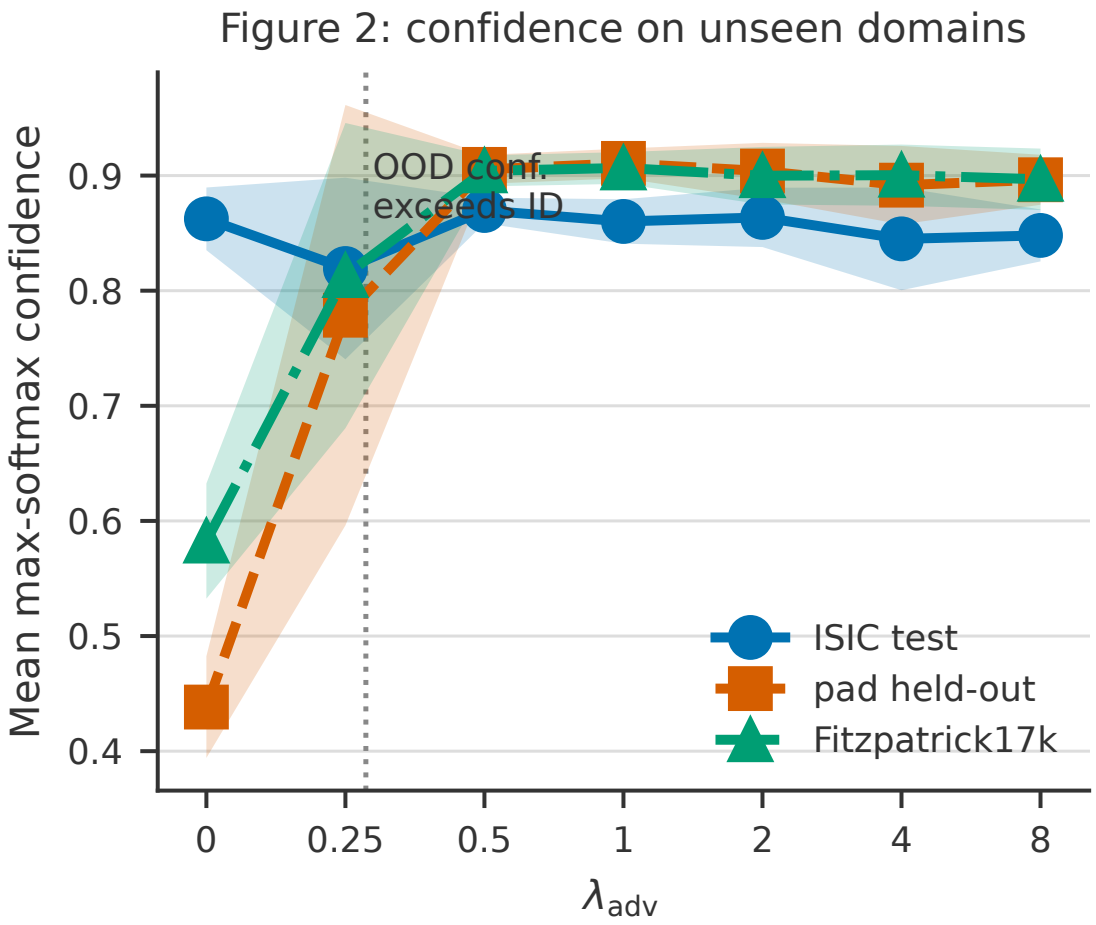


Figure 5: **Confidence on unseen domains overtakes confidence on familiar data.** Mean maximum-softmax probability against λ_{adv} for the ISIC test split, the held-out PAD partition and Fitzpatrick17k. The crossing occurs by $\lambda_{\text{adv}} \approx 0.5$. λ spacing is categorical.

in-distribution set to the out-of-domain class mix raises it to 0.487, so most of that baseline inversion is class composition — `HAM` is two-thirds nevus and therefore concentrated near a single well-populated centroid — with a residual we attribute to HAM10000's more standardised acquisition and correspondingly lower feature variance. This is why the reversed direction, which restores headroom, carries the scope claim in Section 4.2. In that direction the class-reweighted variant falls from 0.74 to 0.54, and Mahalanobis reads 0.719 ± 0.019 at $\lambda_{\text{adv}} = 0.25$.

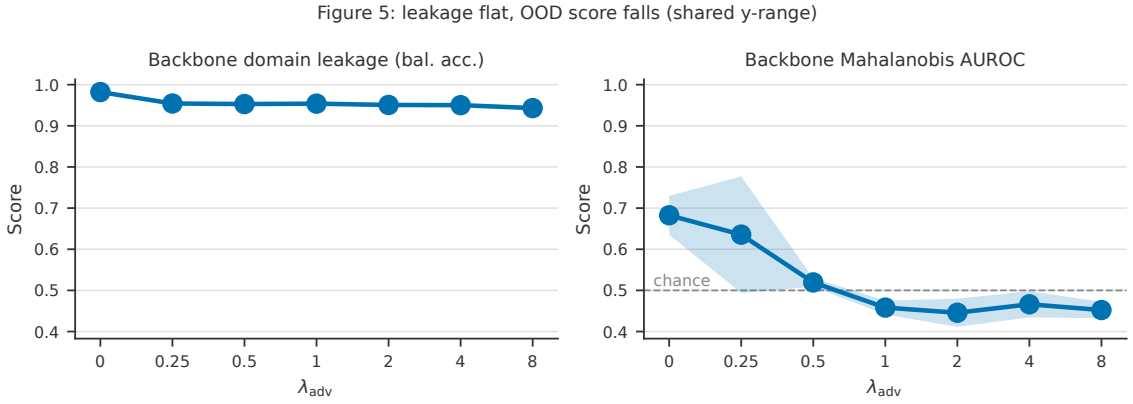


Figure 6: **Leakage does not mediate the collapse.** At the pre-projection backbone the domain probe stays flat across the sweep while Mahalanobis AUROC falls. Domain information remains linearly decodable and detection degrades regardless.

Table 5: Invariance objectives other than gradient reversal, single-encoder backbone, 3 seeds per cell. Leakage is balanced accuracy, floor 0.5. **IRM and GroupDRO use PAD labels** through their environment and group risks; the remaining objectives do not, so their cross-domain numbers are not comparable and the two regimes are kept in separate blocks.

Objective	Weight	Leakage	Maha. AUROC	ID bal. acc	OOD ECE
No access to PAD labels					
ERM[§]	—	0.973	0.838	0.761	—
CORAL	100	0.998	0.999	0.777	—
MMD	1	0.967	0.813	0.779	—
MMD	100	0.900	0.716	0.730	—
MMD	3000	0.893	0.748	0.382	0.142
MMD	10000	0.836	0.665	0.337	0.267
GRL	$\lambda_{adv} = 0.25$	**0.763**	**0.525**	**0.845**[†]	—
Uses PAD labels — not comparable on cross-domain accuracy					
IRM	100	0.950	0.729	0.167	—
GroupDRO	10	0.976	0.868	0.783	—

[†] plain accuracy for the GRL row; see Table 6.
[§] scored on `pad_full` with a uniform optimiser and an ISIC-only loss; the λ_{adv}=0 row of Table 6 uses `pad_heldout`, grouped learning rates and `ignore_index`. They are not the same condition. [PENDING: harmonise or relabel.]

A.6. A domain monitor is cheap, and semantic novelty is unaffected

A linear domain head on a frozen ImageNet ResNet-50 that has never seen this data reaches AUROC 0.998; on the trained baseline, 0.997; on the trained EfficientNet-B3 control, 0.991. Detecting the acquisition domain is easy from almost any representation, so retaining a

Table 6: Single-encoder DANN on the same data and splits. ID accuracy is plain accuracy; leakage is balanced accuracy, floor 0.5. All five detectors collapse at the smallest non-zero weight, and four of five fall below chance by $\lambda_{\text{adv}} = 4$.

λ_{adv}	n	Leakage	Maha.	k-NN	MSP	cosine	Energy	ID acc	X-dom
0	5	0.970	0.700	**0.944**	0.868	0.884	**0.911**	0.845	0.390
0.25	3	**0.763**	0.525	**0.528**	0.515	0.509	**0.507**	0.845	0.333
0.5	3	0.795	0.510	0.517	0.501	0.493	0.507	0.844	0.328
1	3	0.795	0.548	0.521	0.462	0.482	0.448	0.840	0.334
2	5	0.807	0.509	0.516	0.479	0.468	0.470	0.839	0.325
4	3	0.798	0.502	0.486	0.471	0.454	0.449	0.835	0.320
8	3	0.787	0.520	0.532	0.519	0.490	0.500	0.832	0.321

Table 7: Candidate explanations tested and refuted. Each was a prediction specified before its test.

Explanation	Test	Result
Memorisation of the adversarial images	Hold PAD out of the adversary at patient level	`pad_heldout` 0.427 vs `pad_adv` 0.407; unchanged
Class composition	Restrict ID to the six shared classes; reweight ID to the OOD class mix	0.432 vs 0.427 on `pad_heldout`; reweighting *strengthens* the inversion to 0.259
Domain specificity	Evaluate on Fitzpatrick17k, never in $\mathcal{L}_{\text{adv}}$	0.391 at $\lambda_{\text{adv}} = 2$ against 0.764 at $\lambda_{\text{adv}} = 0$, same model and data
Dimensional collapse	Participation ratio and eigenspectrum across λ_{adv}	4.879 $\rightarrow$ 4.500 while leakage 0.914 $\rightarrow$ 0.553 (same three runs); decoupled
Latent compression	Latent variance and distance to centroid across λ_{adv}	OOD is already 4.5$\times$ tighter at $\lambda_{\text{adv}} = 0$; the adversary *decompresses* it
Tail asymmetry	Recompute AUROC after truncating the pooled upper tail	0.412–0.424 versus 0.427; medians invert
Synthetic reproduction	Pre-registered toy model with heavy-tailed ID structure, grid fixed before fitting	No inversion (λ_{adv}=0: 0.693; λ_{adv}=2: 0.786)

monitor is not in itself difficult. What distinguishes the dual-encoder's context branch is *concentration*: a single principal component carries 84.5% of its variance and already separates the domains (AUROC > 0.9999, two discordant pairs in 11.6M), against 0.55–0.70 at $k = 1$ for every other representation tested.

The damage is specific to covariate shift. With the domain held constant and diagnostic classes withheld from training instead, both branches land between 0.6 and 0.8 and the ordering depends on which class is withheld (DF and VASC: lesion 0.648, context 0.641, baseline 0.687; SCC: lesion 0.806, context 0.607, baseline 0.684). No near/far structure organises the difference (Pearson -0.12 against a centroid-similarity index). We report this as a null result.

Table 8: The intervention under the image-level ISIC split, with the two non-adversarial controls. Five seeds at $\lambda_{\text{adv}} \in \{0, 0.25, 1, 2, 8\}$, three at $\lambda_{\text{adv}} \in \{0.5, 4\}$.

λ_{adv}	n	Leakage ↓	ID bal. acc ↑	ID ECE	OOD ECE	OOD AUROC	Fitz. AUROC	X-dom bal. acc
0	5	0.915 ± 0.021	0.692	0.107	0.247	0.843 ± 0.023	0.764 ± 0.038	0.291 ± 0.016
0.25	5	0.571 ± 0.027	0.692	0.100	0.694	0.509 ± 0.023	0.460 ± 0.057	0.295 ± 0.028
0.5	3	0.555 ± 0.029	0.701	0.096	0.724	0.465 ± 0.015	0.449 ± 0.025	0.270 ± 0.034
1	5	0.591 ± 0.035	0.695	0.097	0.735	0.457 ± 0.026	0.414 ± 0.086	0.258 ± 0.012
2	5	0.584 ± 0.027	0.707	0.099	0.746	0.427 ± 0.025	0.391 ± 0.068	0.249 ± 0.030
4	3	0.548 ± 0.035	0.686	0.100	0.726	0.455 ± 0.039	0.431 ± 0.036	0.276 ± 0.025
8	5	0.625 ± 0.045	0.679	0.099	0.713	0.494 ± 0.027	0.474 ± 0.060	0.269 ± 0.023
EffNet-B3 control		0.802	0.683	0.096	—	0.726 ± 0.021	0.649	0.298 ± 0.010
ResNet-50 baseline		0.979	0.655	0.101	—	0.868 ± 0.019	0.994	0.391